

SCRec: Hierarchical Simplicial Modeling for Self-Supervised Recommendation

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Abstract

Graph-based self-supervised learning has emerged as a powerful paradigm for recommendation systems. However, existing methods primarily model pairwise user-item interactions, overlooking the hierarchical compositional semantics inherent in collaborative data. Such semantics arise as items could compose into functional units revealing user preference, while users stratify into communities that shape item characteristics. In this paper, we propose SCRec, a novel framework that introduces simplicial complexes (SCs) into self-supervised recommendation. Unlike standard hypergraph aggregation, which does not enforce nested subset relations, or motif models with a fixed pattern vocabulary, SCs encode multi-scale subset consistency through their downward closure property. Specifically, SCRec couples three designs: (i) recommendation-native simplicial construction from co-consumption patterns, (ii) a vertex-level HiPPR operator that injects higher-order topology without maintaining simplex embeddings during training, and (iii) heterogeneous local-global alignment between collaborative and topological views. Experiments on three real-world datasets demonstrate that SCRec achieves consistently strong recommendation performance against representative pairwise and higher-order baselines, while ablations verify that its structural, HiPPR, and cross-view components are complementary.

1 Introduction

Graph-based recommendation systems have achieved remarkable progress by leveraging graph neural networks (GNNs) to capture collaborative signals (Wang et al. 2019; Chen et al. 2020). Because real-world recommendation scenarios suffer from label scarcity (Liu et al. 2021; Lin et al. 2021), self-supervised learning (SSL) has emerged as a prominent paradigm for extracting informative signals from unlabeled interactions (Veličković et al. 2019; Yu et al. 2022; Cai et al. 2023b; Guo et al. 2024; Zhang et al. 2025; Xia et al. 2023; Lin et al. 2022; Cai et al. 2023a; Yu et al. 2023; Wang et al. 2022). Despite their success, existing SSL frameworks predominantly model pairwise user-item interactions, overlooking the hierarchical compositional semantics inherent in collaborative data, where items compose into functional

units revealing user preference, while users simultaneously stratify into communities that shape item characteristics.

As illustrated in Figure 1, such hierarchical compositional semantics capture rich signals that reshape user-item relationships beyond pairwise interactions. Most existing graph-based SSL methods decompose these structures into independent user-item pairs, overlooking the implicit motivations driving user choices. Consider a concrete example: items such as {yoga mat, kettlebell, dumbbells} collectively form a “home fitness” concept, yet dyadic modeling only captures individual item similarities, failing to recognize that these items function as a coherent unit, which could reflect a user’s lifestyle preference. Extending the co-purchased set to {yoga mat, kettlebell, dumbbells, water bottle} further clarifies why representation choice matters (Figure 1). A hypergraph places all four items in one hyperedge, treating core fitness equipment and an incidental purchase uniformly. A motif-based method may match a predefined triangle over the three fitness items while discarding the water bottle. By downward closure, the core set {yoga mat, kettlebell, dumbbells} forms a coherent 2-simplex whose nodes and edges are all retained, whereas the water bottle remains only weakly connected through lower-order relations. This multi-scale compositional bias distinguishes SCs from both flat hyperedges and fixed motif templates. Similarly, users who purchase this combination may share underlying fitness-oriented profiles, and this community structure in turn reinforces the characteristic of such item groups.

To bridge this gap, hypergraph-based (Wang et al. 2020; Yu et al. 2021; Xia et al. 2022b; Xia, Huang, and Zhang 2022; Ji et al. 2020) and motif-based (Zhang et al. 2025; Zhang and McAuley 2020; Nguyen et al. 2022; Benson, Gleich, and Leskovec 2016) methods capture group-wise interactions beyond dyadic edges. However, incidence-based hypergraph propagation does not structurally enforce consistency between a group and all of its subsets, while motif-based methods restrict higher-order interactions to a predefined pattern vocabulary. In contrast, SCs, rooted in algebraic topology (Spanier and Spanier 1989; Bodnar et al. 2021; Barbarossa and Sardellitti 2020; Schaub et al. 2021), offer a principled framework for modeling hierarchical compositional semantics. The key advantage lies in their downward closure property: any subset of a simplex is itself a simplex. As the example above illustrates, this property preserves a functional

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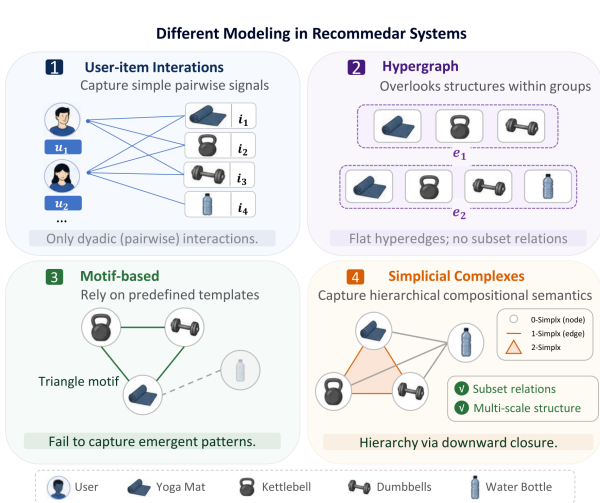


Figure 1: Higher-order structures for recommendation. Pair-wise graphs omit group interactions; standard hypergraph propagation does not enforce subset consistency; motifs use predefined patterns; simplicial complexes explicitly preserve nested faces through downward closure.

unit together with all of its meaningful sub-combinations, enabling preference to be captured at multiple granularities. Analogously, when users stratify into communities based on shared consumption patterns, SCs naturally encode how collective demand reshapes item characteristics.

Based on these observations, we propose SCRec, a self-supervised recommendation framework based on simplicial complexes, to capture hierarchical compositional semantics in collaborative data. Rather than treating graph lifting, propagation, and contrastive learning as independent modules, SCRec couples them in a single pipeline: it constructs recommendation-native SCs from co-consumption patterns, distills their higher-order topology into a vertex-level HiPPR operator designed for ranking, and aligns the resulting global compositional view with a local view of the original bipartite graph. This design preserves the prediction granularity of collaborative filtering while injecting multi-scale structural dependencies into user and item representations. In summary, the contributions of our paper are as follows:

- We identify hierarchical compositional semantics as a structural signal that is not explicitly preserved by pairwise recommendation, standard hypergraph aggregation, or fixed motif templates, and model it through the downward-closure bias of SCs.
- We develop three coupled designs: recommendation-native simplicial construction, a vertex-level HiPPR operator that avoids order-specific simplex embeddings during training, and heterogeneous local–global alignment under recommendation objectives.
- Experiments on three real-world datasets demonstrate competitive performance against strong baselines, while component and sparsity analyses examine where its structural bias is beneficial.

2 Related Work

2.1 Self-Supervised Recommendation

Graph-based SSL recommenders derive supervision from interaction data to alleviate label sparsity. SGL (Wu et al. 2021) contrasts stochastically perturbed interaction graphs, SimGCL (Yu et al. 2022) replaces structural augmentation with embedding perturbations, and LightGCL (Cai et al. 2023b) constructs a global view through low-rank graph signals. IGCL (Guo et al. 2024) and GNNCL (Chen, Zhou, and Ma 2024) further refine informative signals and contrastive objectives. These methods improve representation learning, but their views remain derived primarily from dyadic user-item connectivity; they do not impose a representation in which group interactions and their lower-order subsets remain mutually consistent.

2.2 Higher-Order Recommendation

Higher-order recommenders use hypergraphs or motifs to move beyond pairwise interactions. Hypergraph models, including HyRec (Wang et al. 2020), MHCN (Yu et al. 2021), HCCF (Xia et al. 2022b), SHT (Xia, Huang, and Zhang 2022), and DHCF (Ji et al. 2020), aggregate information through group-wise hyperedges (Antelmi et al. 2023). This captures set-level co-occurrence, but the usual incidence-based representation does not require a hyperedge to contain a hierarchy of consistent sub-interactions. MotifGCN (Zhang et al. 2025) and mixed-order GCN (Zhang and McAuley 2020) instead aggregate instances of predefined motifs, making their structural vocabulary interpretable but fixed before learning. SCs provide a different inductive bias: every simplex contains all of its faces, so group structure is represented jointly at multiple resolutions rather than as an isolated set or a selected pattern.

2.3 Learning on Simplicial Complexes

Simplicial neural models propagate features attached to nodes, edges, and higher-order simplices through boundary or Hodge operators. BScNets (Chen, Gel, and Poor 2022), Hodge-aware convolution (Yang, Leus, and Isufi 2025), and higher-order graph convolution on SCs (Huang et al. 2024) explicitly maintain order-specific simplicial representations. TopoSRL (Madhu and Chepuri 2023) contrasts augmented SCs while preserving simplex-level topology, TopoGCL (Chen, Frías, and Gel 2024) contrasts graph-level persistent representations, and SSN (Madhu, Gurugubelli, and Chepuri 2024) extracts task-agnostic multi-scale scattering features from a predefined SC. These methods target general node, graph, edge-flow, simplex, or text tasks (Raghuvanshi et al. 2024; Liang et al. 2025; Iacopini et al. 2019); directly applying them to collaborative filtering introduces a granularity mismatch because recommendation requires scalable vertex-level user-item scoring on a bipartite graph.

3 Preliminary

3.1 Problem Setting

We consider a standard collaborative filtering (CF) setting with a bipartite user-item interaction graph $G = (U, I, E)$,

where U and I denote the sets of users and items, and $E \subseteq U \times I$ is the set of observed interactions. This graph is represented as a binary matrix $\mathbf{A}$, with $A_{ui} = 1$ indicating an interaction between user u and item i . The objective is to learn user and item embeddings $\mathbf{z}_u$ and $\mathbf{z}_i$ to predict interaction scores $\hat{y}_{ui} = f(\mathbf{z}_u, \mathbf{z}_i)$, which are used to rank items for each user.

3.2 Simplicial Complexes

Simplicial complexes $\mathcal{K}$ generalize graphs to model higher-order interactions beyond dyadic links (Battiston et al. 2020; Benson, Gleich, and Leskovec 2016; Schaub et al. 2021). A k -simplex $\sigma = \{v_0, v_1, \dots, v_k\}$ defines an interaction involving $k + 1$ vertices, encompassing 0-simplex (node), 1-simplex (edge), and 2-simplex (triangle), respectively. By adopting SCs, we can effectively characterize the complex, multi-scale structural patterns within user–item interactions.

4 Method

Overview. SCRec is a three-stage pipeline rather than a collection of parallel modules (Figure 2). The first stage represents hierarchical co-consumption in a form that preserves nested subset relations rather than isolated groups; SCRec lifts co-consumption relations from the bipartite graph into a recommendation-native simplicial complex. The second stage injects this compositional structure into user and item representations while retaining vertex-level scoring; SCRec derives two heterogeneous vertex views, a local collaborative view on the original bipartite graph and a global compositional view through the HiPPR operator induced by the lifted complex. The third stage makes the global view ranking-relevant and transfers compositional signals to the local embeddings used for prediction; SCRec optimizes view-specific BPR objectives together with heterogeneous local–global alignment. The coupling is essential: the lifted complex defines the global operator, and the recommendation objectives make its vertex-level output directly comparable with the local collaborative view.

4.1 Recommendation-Native Simplicial Construction

Pairwise user–item edges alone do not expose which items form a functional unit or how users stratify into preference communities. SCRec therefore needs a structure in which group interactions and their lower-order subsets remain mutually consistent. We first enrich the interaction graph with co-consumption relations and then lift the resulting graph into a downward-closed simplicial complex.

Co-occurrence Graph Augmentation Let $G_u = (U, E_u)$ and $G_i = (I, E_i)$ denote user and item co-occurrence graphs. For two nodes x and y of the same type, we set $\omega_{xy} = |\mathcal{N}(x) \cap \mathcal{N}(y)|$, where $\mathcal{N}(\cdot)$ contains their neighbors in the bipartite graph. We retain pairs with $\omega_{xy} > \eta$ and uniformly sample a proportion ρ to control noise and density:

$$E_u^{(s)} = \text{Sample}_\rho(\{(u_a, u_b) \in E_u \mid \omega_{ab} > \eta\}), \quad (1)$$

with the item-side set $E_i^{(s)}$ constructed analogously. We use $\eta = 3$ and $\rho = 0.01$ in all experiments. The retained co-occurrence density varies by dataset (Figure 3) and directly determines how many cliques enter the lifted complex. The undirected augmented graph is $G_{\text{aug}} = (U \cup I, E_{\text{aug}})$, where E_{aug} contains the original user–item edges in both directions together with $E_u^{(s)}$ and $E_i^{(s)}$.

Clique-Complex Lifting We construct a clique complex $\mathcal{K} = \text{Cl}(G_{\text{aug}})$ up to order p (Schaub et al. 2020; Benson, Gleich, and Leskovec 2016). Every $(k+1)$ -clique is lifted to a k -simplex for $0 \leq k \leq p$, and all of its faces are included by construction. Consequently, a group-level co-consumption pattern is represented together with its constituent nodes, edges, and lower-order subgroups. This downward-closed hierarchy is the structural distinction from an isolated hyper-edge or a selected motif instance, and it supplies the topology that the subsequent propagation stage consumes.

4.2 Heterogeneous Dual-View Propagation

Recommendation requires vertex-level user–item scoring, yet compositional semantics live in higher-order interactions. SCRec therefore separates *what* to preserve (pairwise collaborative evidence vs. global compositional structure) from *how* each signal is propagated at the same prediction granularity. We build a local view on the original bipartite graph and a global view by distilling the lifted complex into a sparse vertex-to-vertex operator, avoiding trainable embeddings on every simplex during optimization.

Local Collaborative View This view preserves direct user–item collaborative evidence and neighborhood-level preference patterns. Let $\mathbf{G}_0^{(u)} = \mathbf{H}_0^{(u)} \in \mathbb{R}^{|U| \times d}$ and $\mathbf{G}_0^{(i)} = \mathbf{H}_0^{(i)} \in \mathbb{R}^{|I| \times d}$ be trainable embeddings. Following simplified graph propagation (Wu et al. 2019; He et al. 2020), layer l is computed as

$$\mathbf{G}_l^{(u)} = \tilde{\mathbf{A}}_l \mathbf{G}_{l-1}^{(i)}, \quad \mathbf{G}_l^{(i)} = \tilde{\mathbf{A}}_l^\top \mathbf{G}_{l-1}^{(u)}, \quad (2)$$

where $\tilde{\mathbf{A}}_l = \mathbf{D}_u^{-1/2} \mathcal{D}_l(\mathbf{A}) \mathbf{D}_i^{-1/2}$ and $\mathcal{D}_l$ denotes layer-wise edge dropout (Rong et al. 2020). The final local representations are $\mathbf{G}^{(u)} = \sum_{l=0}^L \mathbf{G}_l^{(u)}$ and $\mathbf{G}^{(i)} = \sum_{l=0}^L \mathbf{G}_l^{(i)}$.

Global Simplicial View This view injects multi-scale compositional dependencies from the lifted complex into user and item embeddings without introducing order-specific simplex parameters. We design a vertex-level HiPPR (higher-order personalized PageRank) operator that routes information through shared p -simplex membership and then personalizes diffusion back to the vertex space required for ranking. Let $n = |U| + |I|$, N_p be the number of p -simplices, and $\mathbf{B}_p \in \{0, 1\}^{n \times N_p}$ be the vertex–simplex incidence matrix, with $(\mathbf{B}_p)_{v\sigma} = 1$ iff $v \in \sigma$. We first define a row-normalized vertex–simplex–vertex transition that encodes how compositional units mediate communication between vertices:

$$\tilde{\mathbf{A}}^p = \mathbf{D}_v^{-1} \mathbf{B}_p \mathbf{D}_\sigma^{-1} \mathbf{B}_p^\top, \quad (3)$$

where $\mathbf{D}_\sigma$ and $\mathbf{D}_v$ contain simplex sizes and the resulting vertex degrees, respectively; inverse degree entries are set to

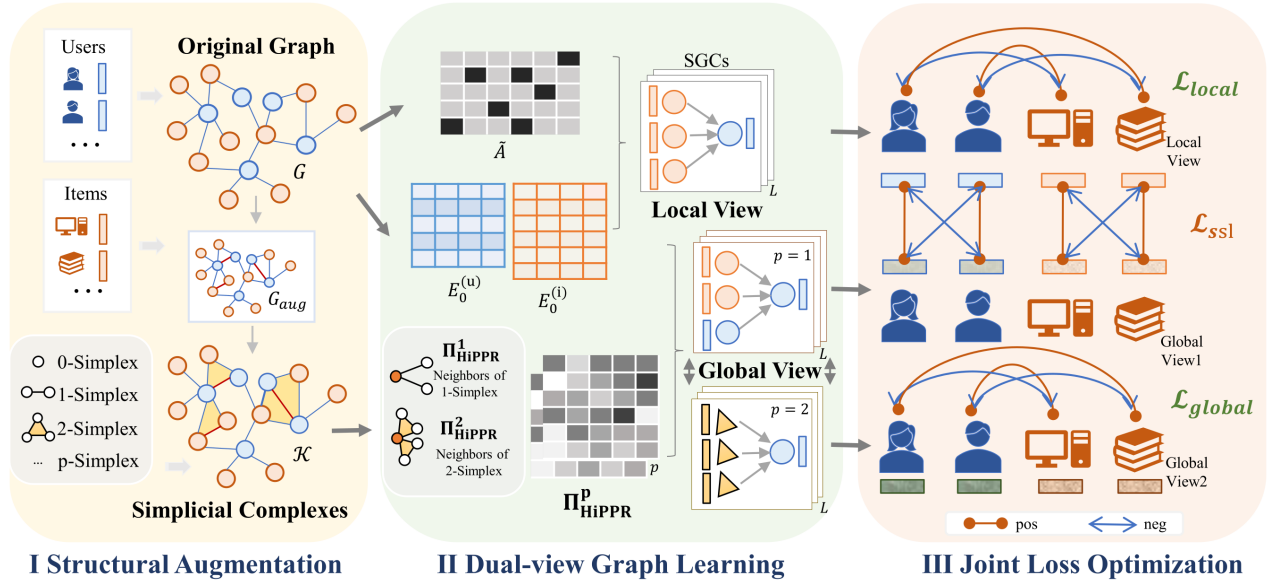


Figure 2: SCRec’s coupled pipeline. (i) Recommendation-native construction lifts co-consumption patterns into a downward-closed SC. (ii) Local propagation preserves pairwise collaborative evidence, while HiPPR produces a global compositional view. (iii) View-specific BPR objectives and local–global alignment jointly optimize both views for recommendation.

zero for vertices not incident to any retained p -simplex. Thus, two vertices exchange information only through membership in a common p -simplex, and the transition remains in the vertex space required for user–item scoring.

We then compose this transition with personalized diffusion to obtain SCRec’s order- p HiPPR operator:

$$\Pi_{HiPPR}^p = \gamma(\mathbf{I}_n - (1 - \gamma)\tilde{\mathbf{A}}^p)^{-1}, \quad (4)$$

where $\mathbf{I}_n$ is the $n \times n$ identity and $\gamma \in (0, 1]$ is the teleport probability. Conceptually, Π_{HiPPR}^p aggregates multi-hop compositional paths while retaining a personalized anchor at each vertex; in practice, we implement it with a sparse local-push scheme (Andersen, Chung, and Lang 2006; Gasteiger, Bojchevski, and Günnemann 2019) and never materialize a dense inverse. Given the concatenated local state $\mathbf{G}_l^{(u+i)} = [(\mathbf{G}_l^{(u)})^\top, (\mathbf{G}_l^{(i)})^\top]^\top$, the global state is

$$\mathbf{Z}_{p,l}^{(u+i)} = \Pi_{HiPPR}^p \mathbf{G}_l^{(u+i)}, \quad \mathbf{Z}_p^{(u+i)} = \sum_{l=0}^L \mathbf{Z}_{p,l}^{(u+i)}. \quad (5)$$

The first $|U|$ rows of $\mathbf{Z}_p^{(u+i)}$ form $\mathbf{Z}_p^{(u)}$, and the remaining rows form $\mathbf{Z}_p^{(i)}$. This design is what allows SCRec to use higher-order topology for recommendation without maintaining a separate simplicial encoder during training.

4.3 Recommendation and Cross-View Objectives

The global compositional view must remain ranking-relevant, and its structural bias should be transferred to the local embeddings used at inference time. We combine view-specific BPR supervision with heterogeneous contrastive alignment across the two vertex views. For view

$x \in \{\text{loc}, \text{glob}\}$, let $s_x(u, i)$ be the inner product between the corresponding user and item representations. Given sampled triplets $\mathcal{B} = \{(u, i, j)\}$ with observed item i and unobserved item j , the two view-specific BPR losses (Rendle et al. 2012) are

$$\mathcal{L}_x = - \sum_{(u,i,j) \in \mathcal{B}} \log \sigma(s_x(u, i) - s_x(u, j)), \quad (6)$$

$$\mathcal{L}_{\text{rec}} = \mathcal{L}_{\text{loc}} + \mathcal{L}_{\text{glob}}.$$

These terms prevent the global view from becoming a purely structural representation disconnected from ranking.

We further align the same user across the global and local views. For the users $\mathcal{B}_U$ in a mini-batch, the layer-wise InfoNCE term is

$$\mathcal{L}_{\text{ssl}}^{(u)} = - \sum_{l=0}^L \sum_{u \in \mathcal{B}_U} \log \frac{\exp(\text{sim}(\mathbf{z}_{p,l}^u, \mathbf{g}_l^u)/\tau)}{\sum_{v \in \mathcal{B}_U} \exp(\text{sim}(\mathbf{z}_{p,l}^u, \mathbf{g}_l^v)/\tau)}, \quad (7)$$

where sim is cosine similarity and τ is the temperature. The item-side loss $\mathcal{L}_{\text{ssl}}^{(i)}$ is defined analogously, and

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda(\mathcal{L}_{\text{ssl}}^{(u)} + \mathcal{L}_{\text{ssl}}^{(i)}). \quad (8)$$

4.4 Complexity Analysis

Clique lifting and HiPPR preprocessing are performed once before training. Their preprocessing cost depends on the retained co-occurrence density and maximum order; SCRec controls this cost through η , ρ , and $p = 2$ rather than constructing unrestricted high-order cliques. Let $m = |E|$ and $m_p = \text{nnz}(\Pi_{HiPPR}^p)$ after sparse approximation. For embedding dimension d , L propagation layers, and batch size B ,

local propagation costs $\mathcal{O}(Lmd)$, global propagation costs $\mathcal{O}(Lm_p d)$, the two BPR terms cost $\mathcal{O}(Bd)$, and in-batch InfoNCE costs $\mathcal{O}(B^2 d)$. The per-iteration training complexity is therefore

$$\mathcal{O}(L(m + m_p)d + B^2 d). \quad (9)$$

Sparse operators and vertex embeddings require $\mathcal{O}(m + m_p + nd)$ memory, excluding the batch similarity matrix. In contrast to simplicial encoders that maintain activations for every order, SCRec does not incur an additional $\mathcal{O}(d \sum_{k=1}^p N_k)$ simplex-embedding term during training.

5 Experiments

We evaluate SCRec through the following research questions:

- **RQ1:** How does SCRec compare with representative pairwise and higher-order recommendation methods?
- **RQ2:** Are simplicial construction, vertex-level HiPPR, and heterogeneous view alignment complementary?
- **RQ3:** How robust is SCRec across users with different interaction frequencies?
- **RQ4:** How sensitive is SCRec to key hyperparameter choices?
- **RQ5:** How efficient and scalable is SCRec under the same evaluation protocol?

We organize the experiments into three layers: *effectiveness* (RQ1–RQ2), *conditional behavior* (RQ3), and *practicality* (RQ4–RQ5, including adapted simplicial SSL baselines; case study in the supplement).

5.1 Experimental Settings

Datasets and Evaluations We evaluate SCRec and baselines on three real-world datasets: **Yelp2018** (Yelp 2018), **Gowalla** (Cho, Myers, and Leskovec 2011), and **Tmall** (IJCAI-15 Competition 2015).

For performance evaluation, we employ Recall@N and Normalized Discounted Cumulative Gain (NDCG@N) (He et al. 2017), with $N \in \{20, 40\}$. We use the same train/validation/test splits as SGL (Wu et al. 2021) and LightGCL (Cai et al. 2023b) and apply the same full-ranking protocol to all methods: each test user ranks all items absent from the training set.

Baselines We evaluate our model against 11 state-of-the-art baselines across diverse learning paradigms in recommender systems:

- **General SSL Baselines:** This category focuses on fundamental paradigms for learning graph representations without manual labels. Early approach GraphCL (You

et al. 2020) uses random node dropping and edge perturbation for graph augmentation and contrastive learning. GCA (Zhu et al. 2021) improves this by applying structure-aware augmentations. SimGRACE (Xia et al. 2022a) eliminates data augmentation by leveraging gradient-based perturbations to generate contrastive signals for better stability.

- **SSL-based Recommendation Systems:** Unlike general SSL methods, these models are specifically designed for recommendation. SGL (Wu et al. 2021) pioneers the integration of structural self-supervised contrastive learning into collaborative filtering. To address the computational overhead and noise sensitivity, SimGCL (Yu et al. 2022) adopts feature-level perturbations, while LightGCL (Cai et al. 2023b) focused on lightweight architectures to balance recommendation accuracy with system efficiency. FGCLR (Wang et al. 2024) further mitigates semantic noise by adopting PCA-based graph augmentation.
- **Higher-Order Recommendation Systems:** To capture complex dependencies beyond simple pairwise interactions, researchers have explored higher-order structures. These include hypergraph recommendation HyRec (Wang et al. 2020), hypergraph contrastive collaborative filtering HCCF (Xia et al. 2022b), hypergraph transformer-based SHT (Xia, Huang, and Zhang 2022) and motif-based recommendation motifGCN (Zhang et al. 2025).

Implementation Details We implement SCRec in PyTorch and conduct the experiments on NVIDIA RTX 3090 GPUs. For fair comparison, baseline hyperparameters are searched within the ranges reported in their original papers under the same data splits and evaluation protocol; SCRec hyperparameters are selected from the ranges below. Shared settings are fixed to an embedding size of 32 and a batch size of 256 when applicable. We train SCRec using AdamW (Loshchilov and Hutter 2019). The contrastive weight λ is selected from $\{0.01, 0.05, 0.1, 0.15\}$, the temperature τ from $\{0.05, 0.1, 0.15, 0.2\}$, and the edge-dropout rate from $\{0, 0.1, 0.2\}$. The maximum SC order is fixed to $p = 2$.

5.2 Main Experiment Results (RQ1)

Table 2 compares general graph SSL methods, recommendation-specific SSL models, and higher-order recommenders on three datasets. SCRec achieves the best results on most dataset–metric combinations and remains competitive on the remaining ones, yielding consistent gains over general SSL methods and several higher-order baselines.

The results support three observations. First, transferring generic graph SSL methods to collaborative filtering is insufficient: recommendation-specific models such as SimGCL, LightGCL, and FGCLR are generally stronger because their learning signals are tailored to user–item data. Second, existing higher-order baselines do not dominate consistently. This pattern is compatible with our structural analysis: hypergraph incidence does not enforce nested subset

Dataset	Users	Items	Interactions
Yelp	29,601	24,734	1,069,128
Gowalla	50,821	57,440	1,172,425
Tmall	47,939	41,390	2,357,450

Table 1: Statistical details of the evaluated datasets.

Dataset	Metric	GraphCL	GCA	SimGRACE	SGL	SimGCL	LightGCL	FGCLR	HyRec	HCCF	SHT	MotifGCN	SCRec
Yelp	R@20	0.0462	0.0621	0.0603	0.0526	0.0718	0.0793	<u>0.0833</u>	0.0472	0.0626	0.0651	0.0777	0.0974
	N@20	0.0401	0.0530	0.0435	0.0444	0.0546	0.0668	<u>0.0707</u>	0.0395	0.0527	0.0546	0.0663	0.0837
	R@40	0.0764	0.1021	0.0989	0.0869	0.1166	0.1292	<u>0.1352</u>	0.0791	0.1040	0.1091	0.1274	0.1551
	N@40	0.0511	0.0677	0.0656	0.0571	0.0778	0.0852	<u>0.0897</u>	0.0522	0.0681	0.0709	0.0846	0.1048
Gowalla	R@20	0.1157	0.0896	0.0869	0.1030	0.1357	0.1578	0.1622	0.0901	0.1070	0.1232	<u>0.1672</u>	0.1926
	N@20	0.0684	0.0537	0.0528	0.0623	0.0818	0.0935	0.0957	0.0498	0.0644	0.0731	<u>0.0994</u>	0.1154
	R@40	0.1904	0.1322	0.1276	0.1500	0.1956	0.2245	0.2293	0.1356	0.1535	0.1804	<u>0.2391</u>	0.2674
	N@40	0.0886	0.0651	0.0637	0.0746	0.0975	0.1108	0.1133	0.0660	0.0767	0.0881	<u>0.1183</u>	0.1351
Tmall	R@20	0.0251	0.0373	0.0222	0.0268	0.0473	0.0528	0.0558	0.0233	0.0314	0.0387	<u>0.0577</u>	0.0834
	N@20	0.0175	0.0252	0.0152	0.0183	0.0328	0.0361	0.0380	0.0160	0.0213	0.0262	<u>0.0401</u>	0.0620
	R@40	0.0416	0.0616	0.0367	0.0446	0.0766	0.0852	0.1235	0.0350	0.0519	0.0645	<u>0.0927</u>	0.0889
	N@40	0.0233	0.0337	0.0203	0.0246	0.0429	0.0473	0.0496	0.0199	0.0284	0.0352	<u>0.0522</u>	0.0742

Table 2: Recommendation performance on three datasets. Best results are in **bold**; second-best are underlined.

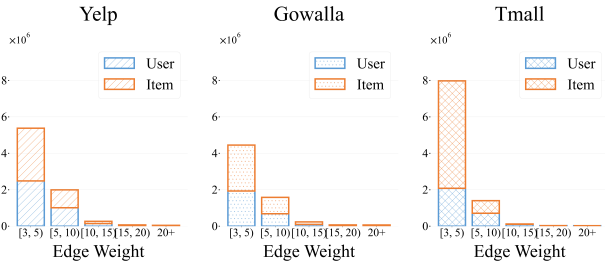


Figure 3: Edge-weight distribution in constructed user-item co-occurrence graphs on Yelp, Gowalla, and Tmall.

consistency, while motif models are restricted to a predefined pattern vocabulary. The accuracy table alone, however, does not establish either limitation as a causal explanation. Third, SCRec’s strong overall performance indicates that its recommendation-native simplicial construction and vertex-level HiPPR can complement local collaborative learning.

Figure 3 summarizes the edge-weight distribution in the constructed user-item co-occurrence graphs under our protocol ($\eta = 3, \rho = 0.01$). Tmall retains the largest volume of weighted co-occurrence edges, whereas Gowalla shows comparatively weak group-level connectivity. This cross-dataset variation is consistent with, though not by itself sufficient to explain, the performance pattern in Table 2: SCRec leads by the widest margin on Yelp, remains competitive with strong recommendation-specific models on Gowalla, and is strongest on top- N ranking metrics on Tmall while FGCLR remains competitive at Recall@40. When co-consumption structure is sparse, the benefit of simplicial lifting depends more on whether local neighborhoods already provide reliable collaborative evidence.

5.3 Ablation Study (RQ2)

We construct three variants to isolate the roles of the coupled design choices. SCRec (**w/o Global**) removes the simplicial global-view objective $\mathcal{L}_{\text{glob}}$. SCRec (**w/o HiPPR**) replaces Π_{HiPPR}^p with a normalized pairwise adjacency operator,

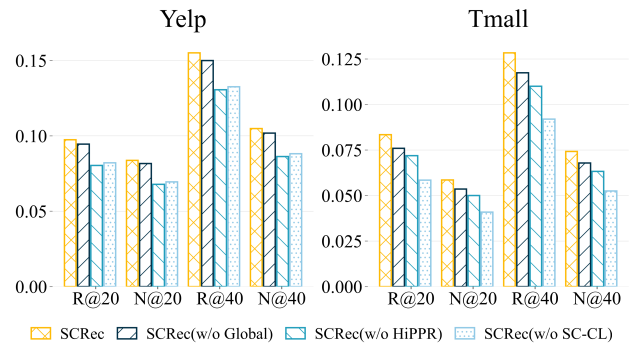


Figure 4: Ablation study on SCRec.

testing whether ordinary graph propagation can substitute for topology-induced HiPPR. SCRec (**w/o SC-CL**) replaces heterogeneous local-global alignment with contrast between two local graph views. As shown in Figure 4, the complete model is strongest in every reported setting. The consistent degradation of all three variants supports complementarity: simplicial construction supplies the structure, HiPPR transfers it to ranking space, and cross-view alignment makes that signal useful to recommendation.

5.4 Effect of Data Sparsity (RQ3)

We partition users by interaction frequency and report Recall@20 for each group on Yelp and Gowalla. As shown in Figure 5, SCRec remains competitive across all observed groups and is particularly effective for low-frequency users, consistent with the HiPPR global view providing complementary signal when local neighborhoods are sparse.

5.5 Hyperparameter Sensitivity (RQ4)

We examine four hyperparameters that directly affect the coupled pipeline: the contrastive weight λ , the temperature τ , the propagation depth L , and the edge-dropout rate used in local propagation. Experiments are conducted on Yelp and

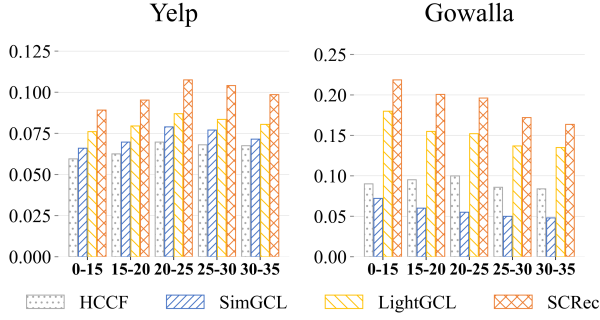


Figure 5: Performance on users of different sparsity degrees, in terms of Recall@20.

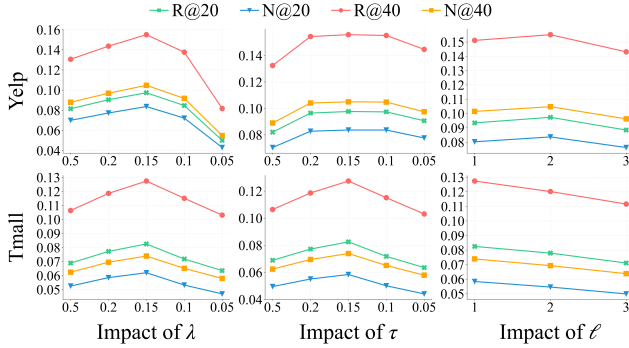


Figure 6: Hyperparameter sensitivity on Yelp and Tmall.

Tmall while keeping the remaining settings fixed to the best values found in the main comparison.

Figure 6 shows that SCRec is not highly brittle within the searched ranges. For the contrastive weight, performance peaks around $\lambda = 0.05$ and remains stable for $\lambda \in \{0.05, 0.1\}$, suggesting that a moderate local-global alignment strength is sufficient without overwhelming the ranking objectives. For the temperature, Recall and NDCG are consistently stronger when $\tau \in [0.1, 0.2]$, indicating that overly sharp contrastive distributions are unnecessary. For propagation depth, using one or two layers provides the best trade-off between representation capacity and stability; deeper propagation does not yield further gains under the current setting. Edge-dropout behaves similarly to prior graph-based recommenders: a small nonzero rate helps regularize local propagation, while excessive dropout removes useful collaborative evidence.

5.6 Efficiency and Scalability (RQ5)

RQ5 complements Table 2 with adapted simplicial SSL baselines and scalability analysis on Yelp. We adapt TopoSRL (Madhu and Chepuri 2023) and SSN (Madhu, Gurugubelli, and Chepuri 2024) to the same protocol on one NVIDIA RTX 3090 GPU (embedding size 32, batch size 256), comparing deployment cost with SCRec and examining training time and peak memory under varying graph scale and retained simplex density.

Metric	TopoSRL [†]	SSN [†]	SCRec
R@20	0.0125	0.0423	0.0974
N@20	0.0104	0.0356	0.0837
R@40	0.0229	0.0713	0.1551
N@40	0.0145	0.0464	0.1048
Time/epoch (s)	23.82	6.95	5.50
Inference (s)	27.76	9.39	8.02
Peak memory (MB)	5439.12	3070.63	2458.00

Table 3: Efficiency and recommendation accuracy on Yelp ([†]adapted simplicial SSL baselines).

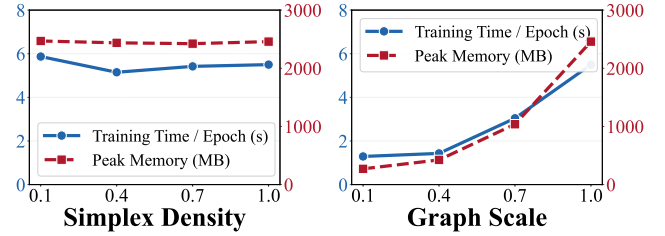


Figure 7: Training time and peak memory on Yelp under varying graph scale and simplex density.

Table 3 shows that SCRec achieves the best recommendation accuracy together with the lowest per-epoch training time and peak memory among the compared methods. Both adapted baselines were tuned with a comparable search budget, yet remain far below SCRec: TopoSRL relies on explicit multi-order simplex embeddings and a symmetric contrastive objective designed for homogeneous node tasks, whereas SSN applies a fixed scattering transform over a predefined complex without end-to-end collaborative learning. Figure 7 further examines scalability on Yelp by subsampling graph scale and varying retained simplex density, where training time grows with graph size and peak memory varies little.

6 Conclusion

We presented SCRec, a self-supervised recommendation framework that encodes hierarchical compositional semantics beyond dyadic interactions. It couples recommendation-native simplicial construction, vertex-level HiPPR, and heterogeneous local-global alignment in a single pipeline, so that compositional structure is encoded, propagated, and aligned for ranking rather than handled by separate modules. Across three datasets, SCRec achieves consistently strong performance relative to representative pairwise and higher-order baselines. Ablations confirm that simplicial construction, HiPPR, and cross-view alignment are complementary. The method also remains effective for low-frequency users when local neighborhoods are sparse, which supports using a global simplicial view together with local collaborative propagation. Compared with adapted simplicial SSL baselines on Yelp, SCRec further achieves better accuracy with lower per-epoch training time and peak memory. Hyperparameter, efficiency, and scalability analyses indicate that SCRec remains practical under standard tuning and hardware settings.

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